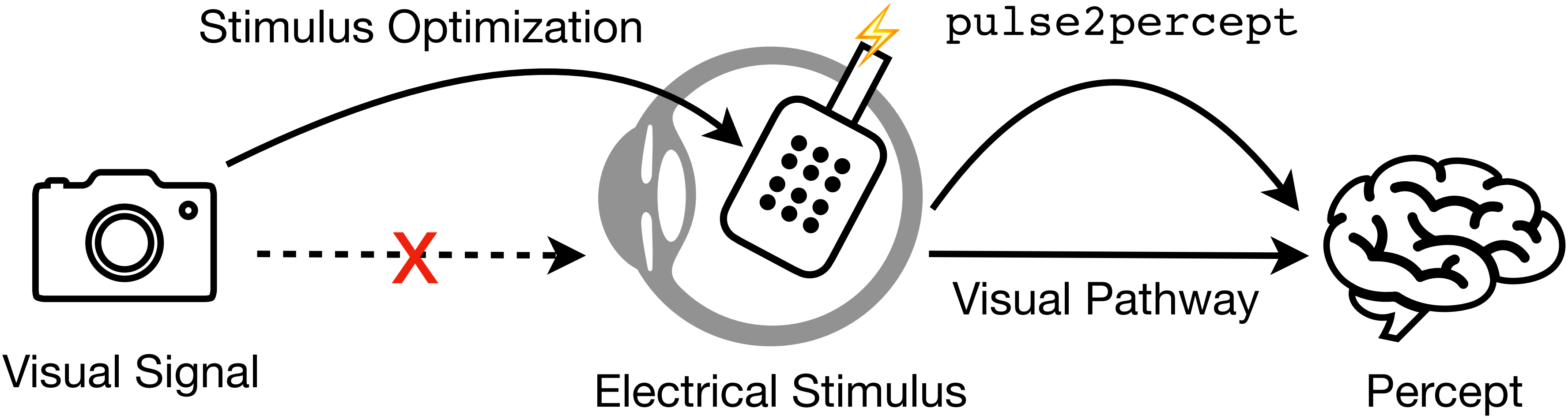


Yuli Multi-Enhancing Artificial Vision by Integrating Machine Learning in Retinal Prostheses

RWTHAACHEN
UNIVERSITY



Problem Formulation



$$\arg \min_{\theta} \mathcal{L}_{eval} \left(\tilde{\mathbf{X}}, f_{\phi} \left(g_{\theta} (\tilde{\mathbf{X}}) \right) \right)$$



$$\mathbf{x}_s = g_{\theta}(\tilde{\mathbf{x}})$$

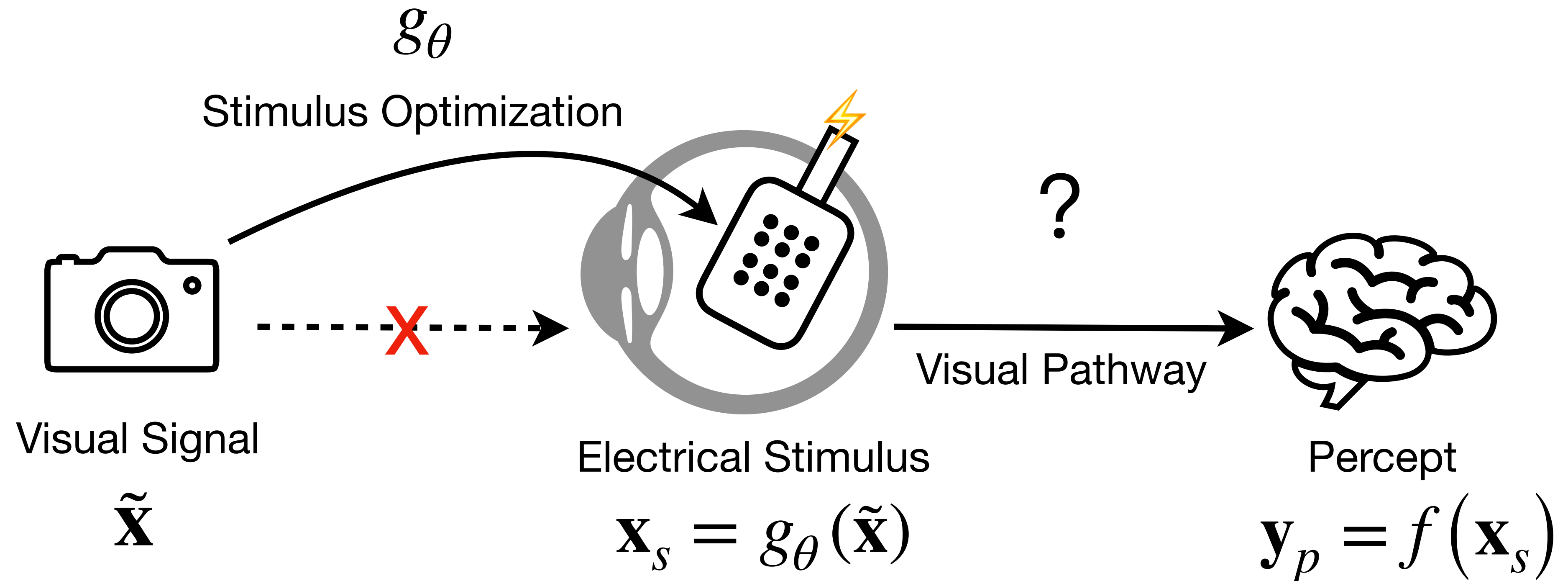
$$\mathbf{y}_p = f(\mathbf{x}_s)$$







Problem Formulation



$$\arg \min_{\theta} \mathcal{L}_{eval} \left(\tilde{\mathbf{x}}, f_{\phi} \left(g_{\theta}(\tilde{\mathbf{x}}) \right) \right)$$

Simulation of Neural Responses